KEDGE BUSINESS SCHOOL · QM1 · 2026–2027

Worksheet 2

Measures of central tendency

Fully worked solutions

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1. Core applications

Master the essential definitions and calculations from the course.

Core application 25 min

Exercise 1 – Choosing and calculating the appropriate measure of centre

1. Basket values are $18, 20, 20, 22, 25, 95$ euros. Calculate mode, median and arithmetic mean. Which measure best represents a typical customer? Why can the mean still matter?
2. Three grades are 14, 11 and 16, weighted by 30%, 30% and 40%. Calculate the weighted mean.
3. Revenue changes by $+5\%$, -2% and $+12\%$. Calculate the appropriate average annual growth rate.
4. A vehicle travels 120 km at 60 km/h and then 120 km at 90 km/h. Calculate the appropriate average speed.

Exercise 1 – Detailed solution

A. Mode, median and mean

$$Mo = 20, \qquad Me = \frac{20+22}{2} = 21,$$
$$\bar{x} = \frac{18+20+20+22+25+95}{6} = 33.33.$$

The median better represents a typical customer because 95 is extreme. The mean is still useful for revenue because $n\bar{x}$ reproduces the total basket value.

B. Weighted mean

$$\bar{x}_w = 14(0.30) + 11(0.30) + 16(0.40) = 13.90.$$

C. Geometric mean

$$1 + g = \sqrt[3]{1.05 \times 0.98 \times 1.12} \approx 1.048442,$$

$$g \approx 4.84\%.$$

D. Harmonic mean

$$H = \frac{2}{\frac{1}{60} + \frac{1}{90}} = 72 \text{ km/h}.$$

Core application 10 min

Exercise 3 – Which indicator should be used?

For each situation, choose the most appropriate indicator and justify in one sentence: most frequently sold product category; typical salary with a few very highly paid executives; average price when purchased quantities differ; average multi-year growth; average speed over equal distances; centre of a roughly symmetric numerical distribution without extreme values.

Exercise 3 – Solution

Situation	Preferred indicator
Most frequently sold category	Mode.
Typical salary with a few extreme high salaries	Median.
Average price with different quantities	Weighted mean.
Average growth over several years	Geometric mean.
Average speed over equal distances	Harmonic mean.
Symmetric numerical distribution without extremes	Arithmetic mean.

Core application 15 min

Extra 1 — Robust centre for delivery performance

Delivery times (minutes) for 12 orders are: 18, 19, 20, 20, 21, 21, 22, 22, 23, 24, 25, 65.

1. Calculate the mode, median and mean.
2. Recalculate the mean without the 65-minute delivery.
3. Recommend the principal indicator for a customer-facing dashboard and justify your choice.

Compare the indicators before and after removing the unusual observation; do not delete it without explanation.

Extra 1 — Robust centre for delivery performance

Detailed solution

Step 1 — mode. The values 20, 21 and 22 each occur twice, so the distribution is multimodal.

Step 2 — median. With 12 observations, use positions 6 and 7: (21 + 22)/2 = **21.5 minutes**.

Step 3 — mean. The total is 300, hence the mean is 300/12 = **25 minutes**. Without the unusual delivery, the total is 235 for 11 orders and the mean is 235/11 = **21.36 minutes**.

Interpretation. A single long delivery raises the mean by 3.64 minutes. The median is the best principal description of a typical customer experience, but the dashboard should also report the mean and the 65-minute incident because the latter matters for capacity and service recovery.

2. Assessment and exam practice

Practise independently with assessment and final-exam style questions.

Assessment / exam practice 30 min

Exercise 2 – Grouped distribution

Class	$[10,14[$	$[14,18[$	$[18,26[$	$[26,30[$	$[30,35[$	$[35,40[$	$[40,45[$	$[45,50[$	$[50,55[$
n_i	4	12	32	32	60	35	25	5	5

1. Identify the modal class, taking unequal widths into account.
2. Calculate the median by linear interpolation.
3. Calculate Q_1 and Q_3 by interpolation.
4. Calculate the approximate mean using class midpoints.
5. Find the maximum price such that about 60% of customers can afford at least that price.

Exercise 2 – Detailed solution

The count densities are $1, 3, 4, 8, 12, 7, 5, 1, 1$, so the modal class is $[30,35[$.

$$Me = 30 + 5\frac{105-80}{60} \approx 32.08.$$

$$Q_1 = 26 + 4\frac{52.5-48}{32} \approx 26.56,$$

$$Q_3 = 35 + 5\frac{157.5-140}{35} = 37.50.$$

Class midpoints are $12, 16, 22, 28, 32.5, 37.5, 42.5, 47.5, 52.5$ and

$$\sum n_i c_i = 6665.$$

Hence:

$$\bar{x} = \frac{6665}{210} \approx 31.74.$$

For 60% to afford at least price p, 40% must lie below it:

$$0.40(210) = 84,$$

$$p = 30 + 5\frac{84 - 80}{60} \approx 30.33 \text{ euros.}$$

Full table for interpolation

Class	n	a	d=n/a	ECC	c	n×c
[10,14[	4	4	1	4	12	48
[14,18[	12	4	3	16	16	192
[18,26[	32	8	4	48	22	704
[26,30[	32	4	8	80	28	896
[30,35[	60	5	12	140	32.5	1950
[35,40[	35	5	7	175	37.5	1312.5
[40,45[	25	5	5	200	42.5	1062.5
[45,50[	5	5	1	205	47.5	237.5
[50,55[	5	5	1	210	52.5	262.5
Σ	210	—	—	—	—	6665

$$Q_1 = 26 + 4\frac{52.5 - 48}{32} = 26.5625, \quad Me = 30 + 5\frac{105 - 80}{60} = 32.0833, \quad Q_3 = 35 + 5\frac{157.5 - 140}{35} = 37.5.$$

Ranks: 0.25×210=52.5; 0.50×210=105; 0.75×210=157.5. Interpolation is approximate because the underlying individual observations are unknown.

Assessment / exam practice 20 min

Extra 2 — Integrated centre-selection case

A retailer reports three indicators:

Question	Data
Average basket value	120 baskets: €3,600 total
Typical annual growth	+12%, −8%, +15%, +5%
Average speed	Two equal 90-km legs at 60 and 90 km/h

1. Choose and calculate the appropriate mean for each row.
2. Explain why using the arithmetic mean in all three cases is incorrect.
3. Write a short managerial conclusion.

Match additive totals, compounded factors and equal-distance rates with arithmetic, geometric and harmonic means.

Extra 2 — Integrated centre-selection case

Detailed solution

Basket value. The arithmetic mean preserves the total: 3,600/120 = **€30**.

Growth. Multiply the growth factors: 1.12 × 0.92 × 1.15 × 1.05 = 1.244208. The equivalent annual factor is $1.244208^{1/4}$ = 1.0561, hence **5.61% per year**.

Speed. Equal distances require the harmonic mean: 2/(1/60 + 1/90) = **72 km/h**.

Why the methods differ. Basket values combine additively, growth compounds multiplicatively, and equal-distance speeds imply unequal travel times. The arithmetic alternatives (6% growth and 75 km/h) do not preserve the relevant totals.

Managerial conclusion. The average basket is €30, the equivalent annual growth rate is about 5.6%, and the route's average speed is 72 km/h. Each result is valid because the averaging rule matches the business mechanism.